

AMAN MISHRA

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TECHNICAL SKILLS

Languages: Python

Frameworks & Libraries: Django Ninja, LangGraph, LangChain, Scikit-learn, SQLAlchemy.

AI / ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic Workflows, Vector Search, Hybrid Retrieval, Reranking, Embeddings

Databases: PostgreSQL

Tools & Concepts: REST APIs, Git, Web Scraping (Crawl4AI), Tavily Search API

AI Observability & Tracing: LangSmith

PROJECTS

Newsletter Agent | 2026

GitHub: <https://github.com/batman00723/Aman-Times-Newsletter.git>

- **Agentic Architecture:** Built a stateful, multi-node workflow using LangGraph to manage complex research and analysis logic.
- **Self-Correction Loop:** Developed a Reflection Node that autonomously critiques drafts against source data to eliminate hallucinations.
- **Async Ingestion:** Leveraged Crawl4AI with asynchronous concurrency to extract clean markdown from diverse, high-impact news sources.
- **Intelligent Filtering:** Implemented an LLM-based scoring rubric to deduplicate and prioritize news on a 1–10 impact scale.
- **Production Backend:** Architected a type-safe API using Django Ninja and Pydantic for automated HTML briefing generation and dispatch.
- **Optimization:** Reduced manual research work into an autonomous pipeline.

RAG-Powered Job Matching Agent | 2026

GitHub: <https://github.com/batman00723/Job-Search-Agent.git>

- Architected a multi-node LangGraph agentic pipeline with intent classification, query rewriting, web scraping, and LLM-based resume-to-job matching.
- Built a reusable RAG pipeline from scratch — custom chunking, embedding generation, hybrid retrieval (vector + keyword), and cross-encoder reranking — shared across multiple agents.
- Integrated Crawl4AI for real-time job page scraping and Tavily for web search with a 3-retry fallback loop on query rewriting.
- Implemented a self-reflection loop where a critique LLM reviews match reports for hallucinations and scoring accuracy before returning results.
- Persisted multi-turn conversation memory using PostgreSQL checkpointing via LangGraph's PostgresSaver.

Credit Card Default Risk Modelling | 2025

GitHub: https://github.com/batman00723/credit_card_default_ml.git

- Engineered behavioural risk signals (delinquency recency, payment discipline, utilization) from 30K records; excluded demographic features like gender to eliminate bias.
- Benchmarked 3 models; rejected Gradient Boosting despite equal ROC-AUC due to validation instability selected constrained Random Forest for generalization reliability.

EDUCATION

B.E. Computer Science & Engineering | Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal

Expected Graduation: 2027